

# Xingyi (Daniel) Chen

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## RESEARCH INTEREST

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Statistical genomics and biomedical data science; applied statistics and machine learning for medicine and healthcare.

## EDUCATION

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### Johns Hopkins University

Baltimore, MD

*Bachelor of Science*

Expected May 2027

- Major: Applied Mathematics & Statistics. Minors: Computational Medicine, Mathematics.
- Cumulative GPA: 3.99. Dean's List Recipient (6 semesters).
- Relevant Coursework:

**Mathematics & Statistics:** Multivariable Calculus, Differential Equations, Linear Algebra, Real Analysis I/II (with Measure Theory) (A+), Probability, Mathematical Statistics (A+), Applied Statistics & Data Analysis (A+), Data Science (A+), Optimization, Monte Carlo Methods (A+), Computational Mathematics, Causal Inference (audit)

**Computational Biology & Genomics:** Computational Biology (A+), Computational Medicine (A+), Computational Genomics: Data Analysis (A+), Genomic Data Visualization, Systems Pharmacology, Healthcare AI Evaluation, Genomics Sequence Data Analysis, Genetics

## RESEARCH EXPERIENCE

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### Research Assistant – Hicks Lab

January 2025 – Present

Johns Hopkins University, Department of Biostatistics | PI: Dr. Stephanie Hicks

- Designed isoform-level machine learning models for human brain aging using bulk RNA-seq, identifying tissue-agnostic and hippocampus-specific transcript usage markers through feature selection and cross-dataset validation; presented findings at Biology of Genomes, CSHL, 2026.
- Developed 10+ reusable Jupyter pipelines and a pip-installable *visiumhd-utils* package for Visium HD preprocessing, QC, destripping, visualization, and publication-ready figure generation for a NIH R01 grant proposal.
- Ported SpotSweeper from R/Bioconductor to Python, preserving spatially aware quality-control functionality and contributing to a published F1000Research software workflow.
- Benchmarked GeneCover against scRNA-seq gene panel selection methods for Xenium panel design, assessing whether compact gene sets preserve biological coverage, marker diversity, cell-type separability, and downstream classification performance.

### QSURE Intern – Shah Lab

New York, NY

Memorial Sloan Kettering Cancer Center | PI: Dr. Sohrab Shah; Mentors: Dr. Andrew McPherson, Matt Myers

Summer 2026

- Developed computational methods to detect homozygous deletions from scDNA data.

### Independent Research Project – scRNA-seq Cell Type Annotation Evaluation Benchmarking

March 2026 – Present

Johns Hopkins University | PI: Dr. Alexis Battle

- Benchmarked scRNA-seq cell type annotation under random cell-level, donor-held-out, and cross-site evaluation schemes using healthy PBMC datasets, demonstrating systematic performance inflation from donor-level leakage.
- Evaluated multinomial logistic regression classifiers across HVG, PCA, Harmony, and scVI representations using macro F1, per-class error analysis, confusion matrices, and transfer benchmarking.
- Performed donor ablation and cell-type-specific analyses, showing that classification difficulty is driven by biological structure and that T-cell subtype generalization remains challenging across representations.
- Developed a fully reproducible benchmarking framework with modular Python pipelines, ordered Jupyter notebooks, manuscript drafts, presentation materials, GitHub repository, and Zenodo archival release.

### Research Assistant – Beer Lab

November 2023 – January 2025

Johns Hopkins University, Department of Biomedical Engineering | PI: Dr. Michael Beer

- Performed exploratory analysis & visualization for gene expression datasets (R: tidyverse, ggplot2)
- Programmed dataset extraction from databases and assisted with downstream data cleaning and processing.

## Bioinformatics Intern – Wei Lab

Lin Gang Laboratory | PI: Dr. Wu Wei

Shanghai, China

Summer 2024

- Synthesized 5 spatial-omics papers into internal talks; highlighted ML/statistics applications & proposed follow-ups.
- Supported 2 in-progress bioinformatics projects with data cleaning and preliminary analysis.

## MANUSCRIPTS & PRESENTATIONS

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[Spotsweeper-py](#) – spatially-aware quality control metrics for spatial omics data in the Python ecosystem.

Chen X, Totty M, and Hicks SC. *F1000Research*. 2026; 15:33. DOI: [10.12688/f1000research.175162.1](https://doi.org/10.12688/f1000research.175162.1)

Chen X, Erdogdu B, Pertea M, Hicks SC. *Machine learning reveals tissue-agnostic and region-specific isoform aging markers in the human hippocampus*. Poster presentation at *Biology of Genomes*, Cold Spring Harbor Laboratory, May 2026.

Chen X, Li J, Battle A. *How much do random cell-level splits overestimate scRNA-seq cell-type annotation performance? A donor-aware multi-cohort benchmark across representations and transfer settings*. Manuscript in preparation.

## SELECTED COURSE RESEARCH PROJECTS

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**BRCA-Mutant Breast Cancer scRNA-seq Analysis** – Semester Project, Foundations of Computational Biology

Johns Hopkins University | Mentor: Dr. Chris Bradburne

January 2025 – May 2025

- Led differential expression part of scRNA-seq analysis: compared gene expression across defined sample clades on custom-built hierarchical trees with dual statistical tests (t-test and U-test) and produced publication ready volcano plots in Python (Scanpy).
- Discovered 6 candidate driver genes whose up or down regulation tracks BRCA-mutant tumor evolution: verified robustness and quantified effects using fold change metric and consensus voting.
- Authored the discussion section of the manuscript, highlighting biological significance while outlining future work.

**Visium Spatial Transcriptomics Dataset Comparison Analysis** – Final Project, Genomic Data Visualization

Johns Hopkins University | Mentor: Dr. Jean Fan

January 2025 – March 2025

- Analyzed Visium spatial transcriptomics data from two kidney tissue sections post-injury using R.
- Applied deconvolution, dimensionality reduction (t-SNE, PCA), and k-means clustering to identify cell-type specific responses at 4 hours vs 2 days after acute kidney injury.
- Identified and interpreted 5 significant marker genes, highlighting spatial patterns and their significance with respect to kidney recovery mechanisms.

**COVID-19 Regression Analysis** – Semester Project, Freshman Experience in Applied Mathematics & Statistics

Johns Hopkins University | Mentor: Dr. Prajakta Bedekar

September 2023 – December 2023

- Conducted regression analysis for COVID-19 antibody levels after vaccination and compared age subgroup results: wrote a 20-page summary paper highlighting methodologies, key results, and future work.

## LEADERSHIP EXPERIENCE

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**Undergraduate Lead Teaching Assistant**

August 2024 – Present

Johns Hopkins University, Department of Mathematics | Supported Course: Differential Equations & Applications, Calculus III

- Promoted from Course Assistant to Lead TA; led weekly discussion section for 25 students and delivered 13 lectures every semester using self-prepared notes.
- Mentored undergraduate TAs, supported course logistics, co-developed quizzes/review sessions, and earned a 98% average student rating.

## SKILLS

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Programming: Python (proficient), R (fluent), SQL, C/C++, Java

Libraries: Scanpy, AnnData, pandas, numpy, matplotlib; ggplot2, tidyverse; scikit-learn

Data/ML: EDA, regression/classification, clustering, reproducible workflows

Technical: Unix, Git/GitHub, LaTeX.

Languages: Mandarin (native), English (bilingual), French (elementary)